

- ≡ Hide menu
- Simpson's Paradox, Weighting, & CMH

Case Control Sampling and Exact Inference for OR

2x2 Tables

Sign, Sign Rank, and Rank Sum

Poisson

Review

Practice Quiz: Module 4 Homework

Started

Quiz: Module 4 Quiz

30 min

🎉 Congratulations! You passed!

Grade received 100% To pass 80% or higher

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Module 4 Homework

1. A case-control study of esophageal cancer was performed. Daily alcohol consumption was ascertained (H = high, L = low). The data was stratified by age group. Ready to review what you've learned before taking the quiz? I'm here to help.

1 / 1 point

Low age group

Help me practice

Let's chat

H L
Case 8 5
Control 52 164

Submit your assignment

Try again

High age group

H L
Case 2521
Control 28 128

Receive grade

Your grade

100%

View Feedback

To Pass 80% or higher We keep your highest score

Give an estimate of the common odds ratio across age strata comparing case status to alcohol usage.

- ☐ The estimate is around 3, suggesting that the estimated age adjusted odds of esophageal cancer are three times higher among the high alcohol users compared to low alcohol users
- ☐ The estimate is around 1, suggesting that the estimated age adjusted odds of esophageal cancer of the high alcohol users are one third that of the low alcohol users.
- ☒ The estimate is around 5, suggesting that the estimated age adjusted odds of esophageal cancer are five times higher among the high alcohol users compared to low alcohol users
- ☐ The estimate is around 5, suggesting that the estimated age adjusted odds of esophageal cancer of the high alcohol users are one fifth that of the low alcohol users

Correct

We will use the Mantel Haenszel estimator

```
1 dat <- array(c(8, 52, 5, 164, 25, 29, 21, 128), c(2, 2, 2))
2 dat
```

```
1 , , 1
2
3      [,1] [,2]
4 [1,] 8    5
5 [2,] 52   164
6
7 , , 2
8
9      [,1] [,2]
10 [1,] 25   21
11 [2,] 29   128
```

```
1 mantelhaen.test(dat, correct = FALSE)
```

```
1 Mantel-Haenszel chi-squared test without continuity correction
2
3 data: dat
4 Mantel-Haenszel X-squared = 32.18, df = 1, p-value = 1.484e-08
5 Alternative hypothesis: true common odds ratio is not equal to 1
6 95 percent confidence interval:
7  2.845 5.592
8 sample estimates:
9 common odds ratio
10  5.197
```

2. The Biostatistics and Epidemiology departments are running a 10K road race. There are three pairs of runners. In each case, a runner from Biostat was matched to a runner from Epi of the same age, gender and degree of running experience. For the pair with the largest difference in their finishing times, the winner was from Epi, for the pair with the second largest difference in their times, the winner was from Biostat and for the final pair, the winner was from Epi.

1 / 1 point

Ranking differences in times from least to greatest and assuming that the times were subtracted as (Epi - Biostat), what would the observed signed rank statistic (sum of the positive ranks) be?

- ☐ The signed rank statistic is 1
- ☒ The signed rank statistic is 2
- ☐ The signed rank statistic is 3
- ☐ The signed rank statistic is 5

Correct

Recall we are ranking from smallest time difference to largest time difference. If the order of subtraction was Epi - Biostat, if the winner of the pair was from Epi, the difference would be negative (a lower time for the Epi runner)

Sign Rank

-	3
-	2
-	1

Thus the observed signed rank statistic is 2.

3. Two computer based methods for diagnostic imaging are being studied. Ten images were processed with both methods, resulting in a + or - diagnosis for each method and image. The data are as follows

1 / 1 point

```
++
+54
-10
```

Where Method A is represented with the two columns and Method B is represented on the rows. What is the P-value for an exact test of the hypothesis that the marginal probability of a positive diagnosis is the same for the two methods?

- ☒ The P-value is around 0.38, we conclude that there is not sufficient evidence to suggest that the marginal probabilities are different
- ☐ The P-value is small (<0.05), we conclude that not sufficient evidence to suggest that the marginal probabilities are different
- ☐ The P-value is small (<0.05), we conclude that there is evidence to suggest that the marginal probabilities are different.
- ☐ The P-value is around 0.19, we conclude that there is not sufficient evidence to suggest that the marginal probabilities are different

Correct

We use a two sided exact McNemar's test

```
1 2 * poisson(3, 5, 0.5, lower.tail = FALSE)
```

```
1 [1] 0.375
```

4. A matched case control study was executed to study an airborne environmental toxicant's effect on lung cancer. The data are

1 / 1 point

Case and exposure status	count
Case exposed, control exposed	243
Case exposed, control unexposed	189
Case unexposed, control exposed	54
Case unexposed, control unexposed	153

What is the marginal odds ratio of odds of exposure for cases (numerator) over odds of exposure for controls (denominator)?

- ☐ Around 0.42
- ☐ Around 5.0
- ☒ Around 2.4
- ☐ Around 3.5

Correct

Solution

```
1 expCase <- 243 + 189
2 unexpCase <- 54 + 153
3 expCtrl <- 243 + 54
4 unexpCtrl <- 189 + 153
5
6 ## marginal odds ratio
7 expCase * unexpCtrl / (unexpCase * expCtrl)
```

```
1 [1] 2.403
```

5. A matched case control study was executed to study an airborne environmental toxicant's effect on lung cancer. The data are

1 / 1 point

Case and exposure status	count
Case exposed, control exposed	243
Case exposed, control unexposed	189
Case unexposed, control exposed	54
Case unexposed, control unexposed	153

Give a large sample P-value for a hypothesis test that the proportion of exposed was the same for the cases and controls

- ☐ The p-value is extremely small and suggesting that the marginal proportions are the same
- ☐ The p-value is modest, suggesting that the marginal probabilities are the same
- ☒ The p-value is extremely small, suggesting that the marginal proportions are different
- ☐ The p-value is modest, suggesting that the marginal probabilities are different

Correct

Solution, we use the large sample McNemar's test

```
1 mcnemar.test(matrix(c(243, 189, 54, 153), 2), correct = FALSE)
```

```
1 McNemar's chi-squared test
2
3 data: matrix(c(243, 189, 54, 153), 2)
4 McNemar's chi-squared = 79, df = 1, p-value < 2.2e-16
```

6. A topical rash treatment was applied to a portion of a rash on n patients. A quantitative measure of redness was calculated for the treated and untreated regions of the rash. A sign of + was given when the treated area was less red (more improved) than the untreated area and a - sign when it was not. It is desired to know whether the treatment improves the rash.

1 / 1 point

For what size of n for a 5% one sided sign test is there a non empty rejection region?

- ☐ You need 5 patients, most have to be +
- ☐ You need 20 patients, all have to be +
- ☒ You need 5 patients and all have to be +
- ☐ You need 20 patients, most have to be +

Correct

```
3 1 / 2^n-1/(2^(1-1/5))
```

```
1 [1] 0.50000 0.25000 0.12500 0.06250 0.03125
```

```
1 ## Thus, we can not reject at the 5% level until we have five
2 ## subjects, and then all five have to be +
```

7. A topical rash treatment was applied to a portion of a rash on n patients. A quantitative measure of redness was calculated for the treated and untreated regions of the rash. A sign of + was given when the treated area was less red (more improved) than the untreated area and a - sign when it was not. It is desired to know whether the treatment improves the rash.

1 / 1 point

Consider a 5% one sided sign test for 5 subjects. What would be the power of the test if the probability that the